



Cluster : Security in distributed wireless networks **(IoT/CPS, 5G, etc.)**

Problem Statement title : Wireless Protocol Fuzzing

Description:

Develop an automated fuzzing tool to discover protocol-level vulnerabilities in distributed wireless systems such as IoT, CPS, Wi-Fi, Zigbee, Bluetooth, and 5G. The tool should identify flaws in protocol handshakes, reconnection flows, and state transitions, enabling early discovery of weaknesses. The goal is to build a scalable, reproducible, and coverage-guided fuzzer that strengthens the resilience of emerging IoT and wireless infrastructure.

Exact Deliverables :

- A working fuzzer for at least one protocol stack (WPA2/3, BLE, Zigbee, or Wi-Fi), with extendable architecture for IoT protocols.
- Automated crash detection and unique bug identification module
- Reporting framework to measure unique crashes and CVEs found against baselines.
- Proof-of-concept demonstrating fuzzing against simulated or open-source protocol implementations.

Milestones, Evolution Parameters:

- **Phase 1:** Implement baseline protocol fuzzing for WPA2/3 or BLE.
- **Phase 2:** Add instrumentation and state coverage tracking.
- **Phase 3:** Extend framework to IoT protocols (Zigbee, LoRaWAN, Matter).
- **Phase 4:** Support multi-protocol fuzzing and real-time reporting.

Additional Information:

- Teams should focus on lightweight fuzzers that can run on commodity hardware or cloud instances.
- Ethical guidelines must be followed: fuzzing should target lab environments, not live production networks.
- Consider modular design so that the fuzzer can be extended to future protocols (e.g., Matter, LoRaWAN).